

Representation in Latent Space Generation



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谢赛宁新作：VAE退役，RAE当立



RAE Public

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Code



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README



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Diffusion Transformers with Representation Autoencoders (RAE)

Official PyTorch Implementation

About

Official PyTorch Implementation of "Diffusion Transformers with Representation Autoencoders"

Readme

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Activity

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Report repository

Releases

No releases published

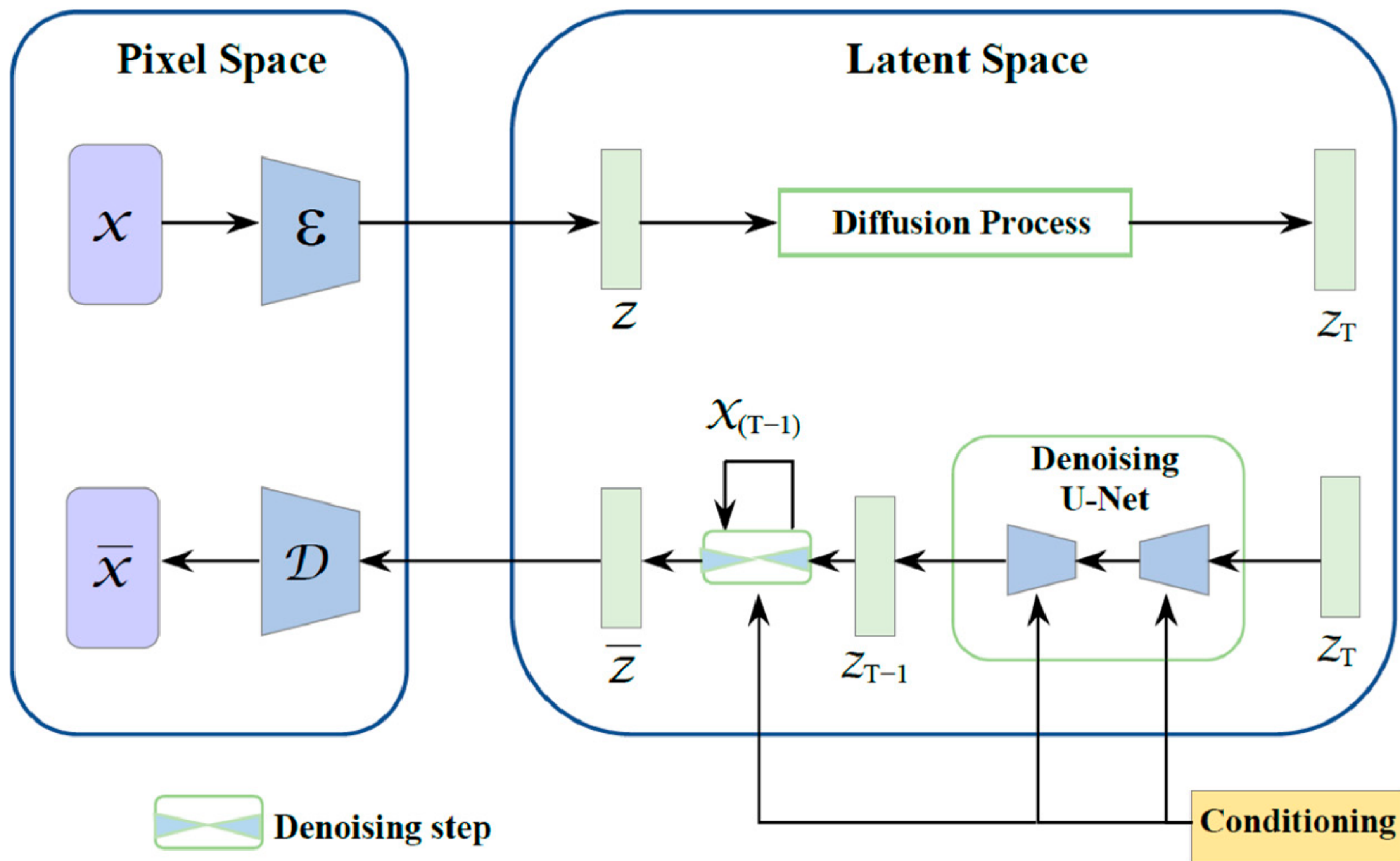
Packages

No packages published

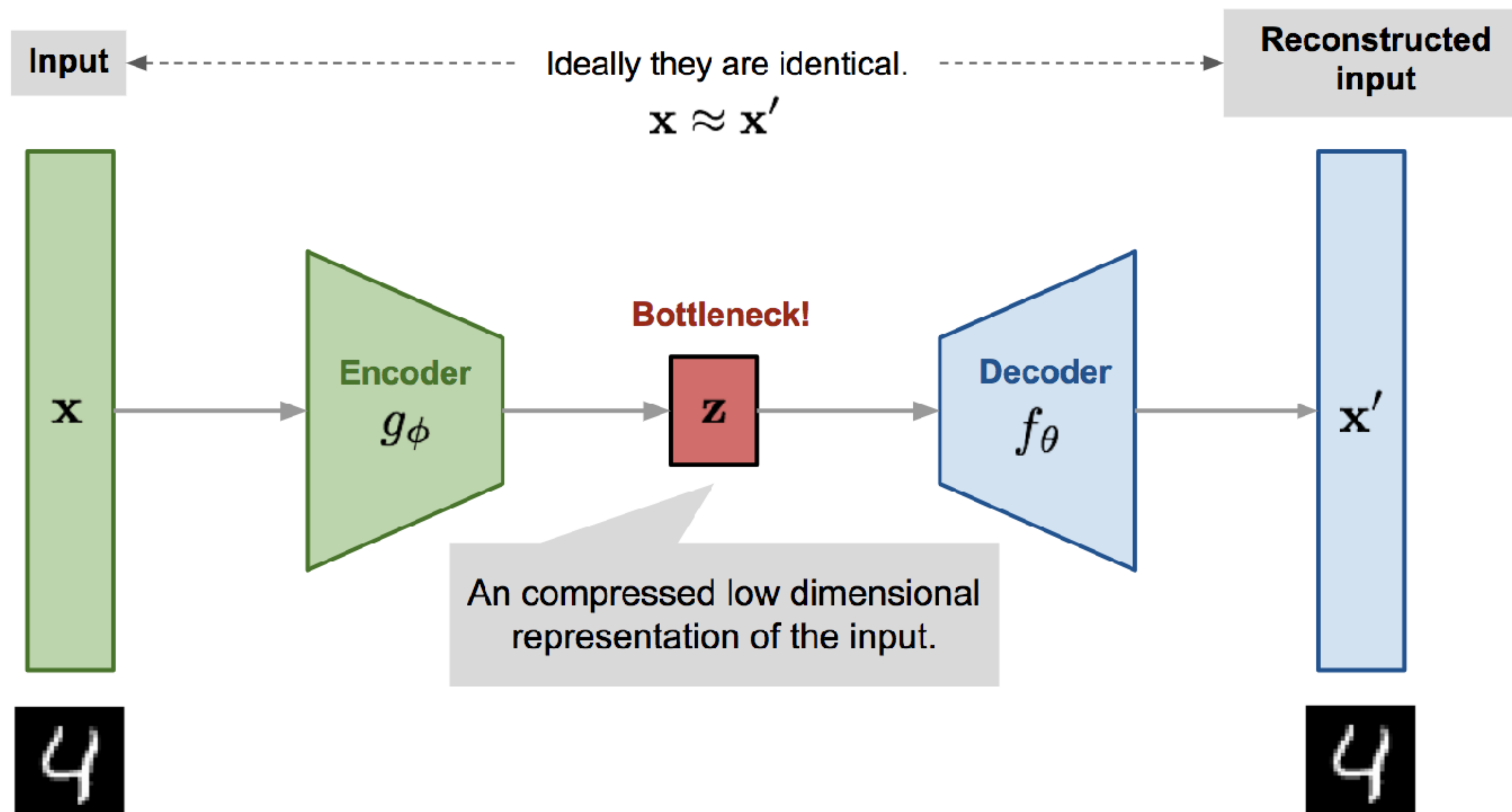
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Latent Space Generation

Latent Diffusion Model



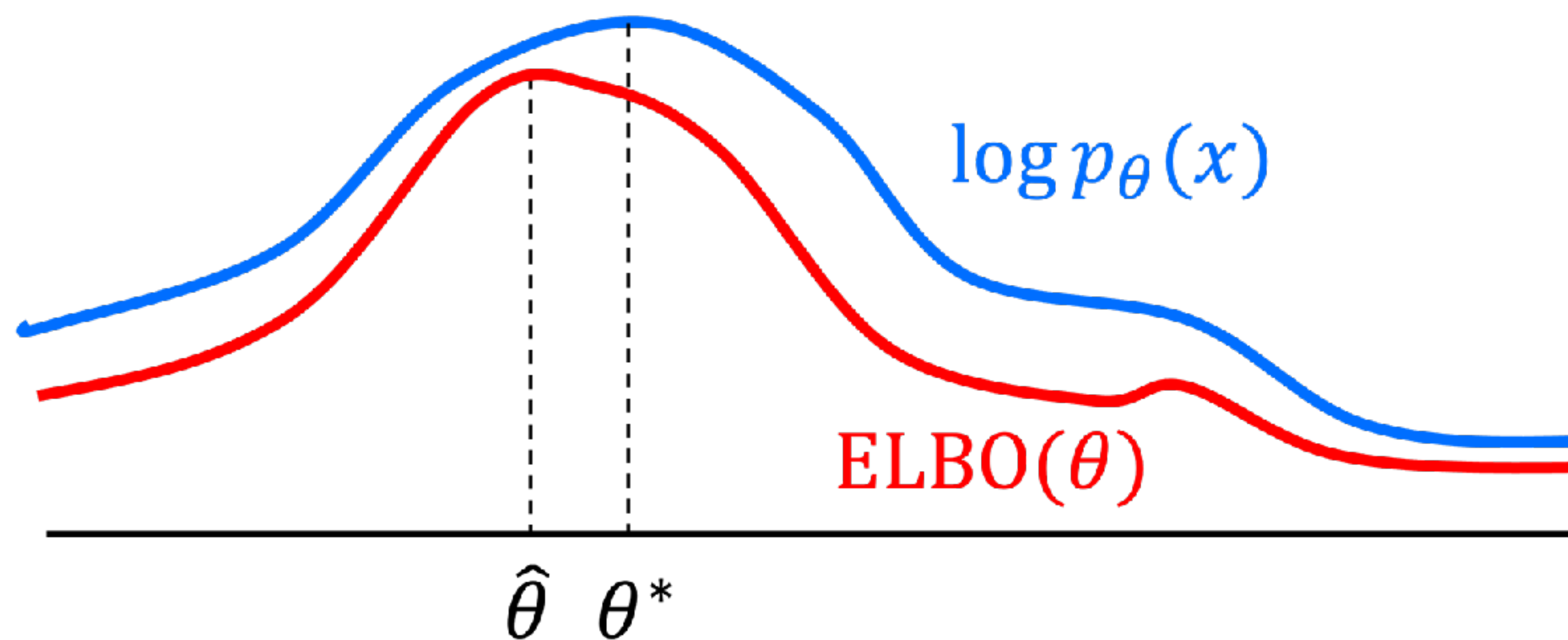
VAE



VAE

ELBO

- Reconstruction+ β *KL
- $\beta=10e-6$

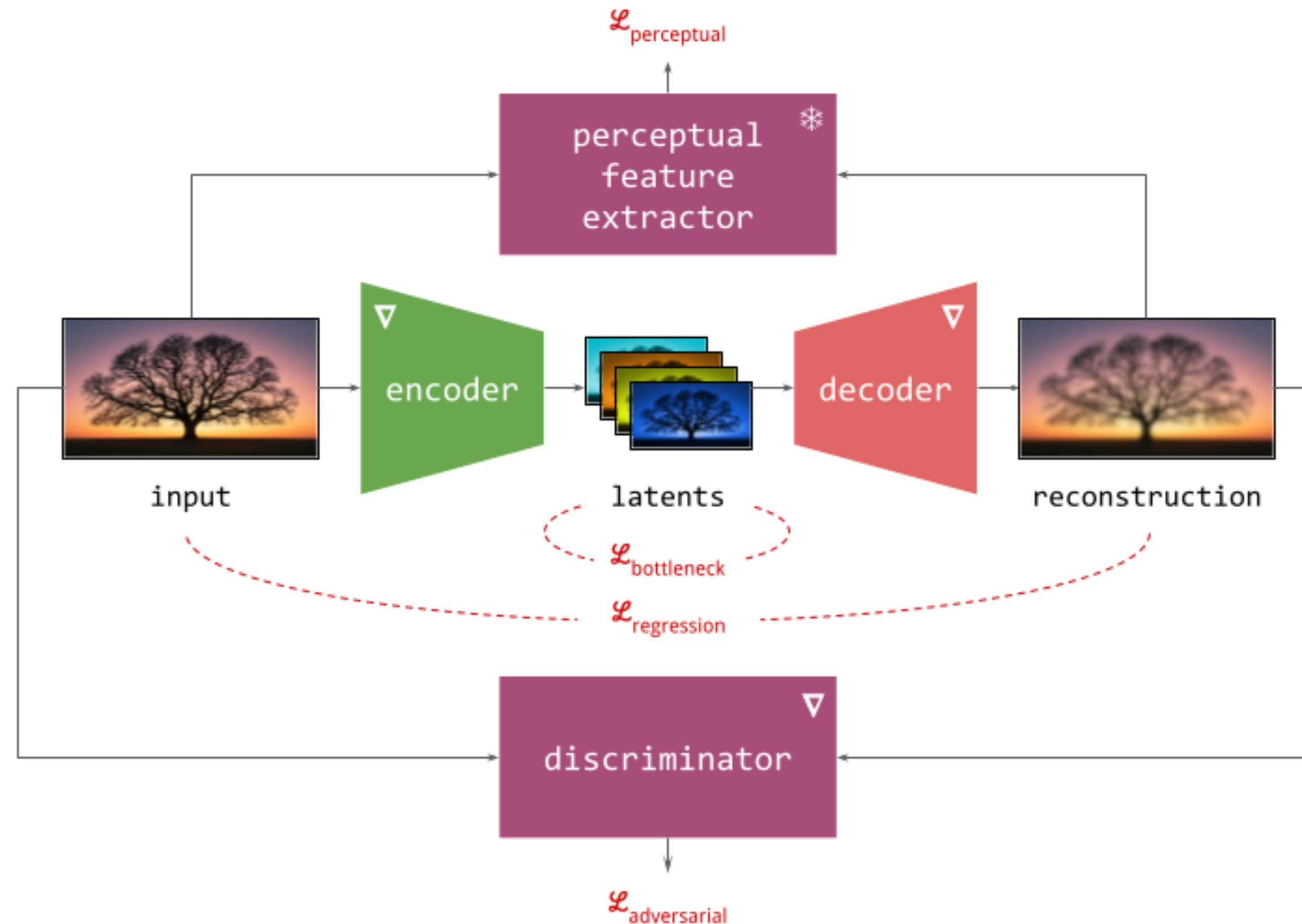


$$\begin{aligned} & \ln p(x) \\ &= \ln \int_z p(x, z) \\ &= \ln \int_z p(x, z) \frac{q(z|x)}{q(z|x)} \\ &\geq \mathbb{E}_{q(z|x)} \left[\ln \frac{p(x, z)}{q(z|x)} \right] \\ &= \mathbb{E}_{q(z|x)} \left[\ln \frac{p(x|z)p(z)}{q(z|x)} \right] \\ &= \mathbb{E}_{q(z|x)} [\ln p(x|z)] + \mathbb{E}_{q(z|x)} \left[\ln \frac{p(z)}{q(z|x)} \right] \\ &= \mathbb{E}_{q(z|x)} [\ln p(x|z)] + \int_z q(z|x) \ln \frac{p(z)}{q(z|x)} \\ &= \mathbb{E}_{q(z|x)} [\ln p(x|z)] - D_{KL}[q(z|x) || p(z)] \\ &= \text{likelihood} - KL \end{aligned}$$

VAE

Loss

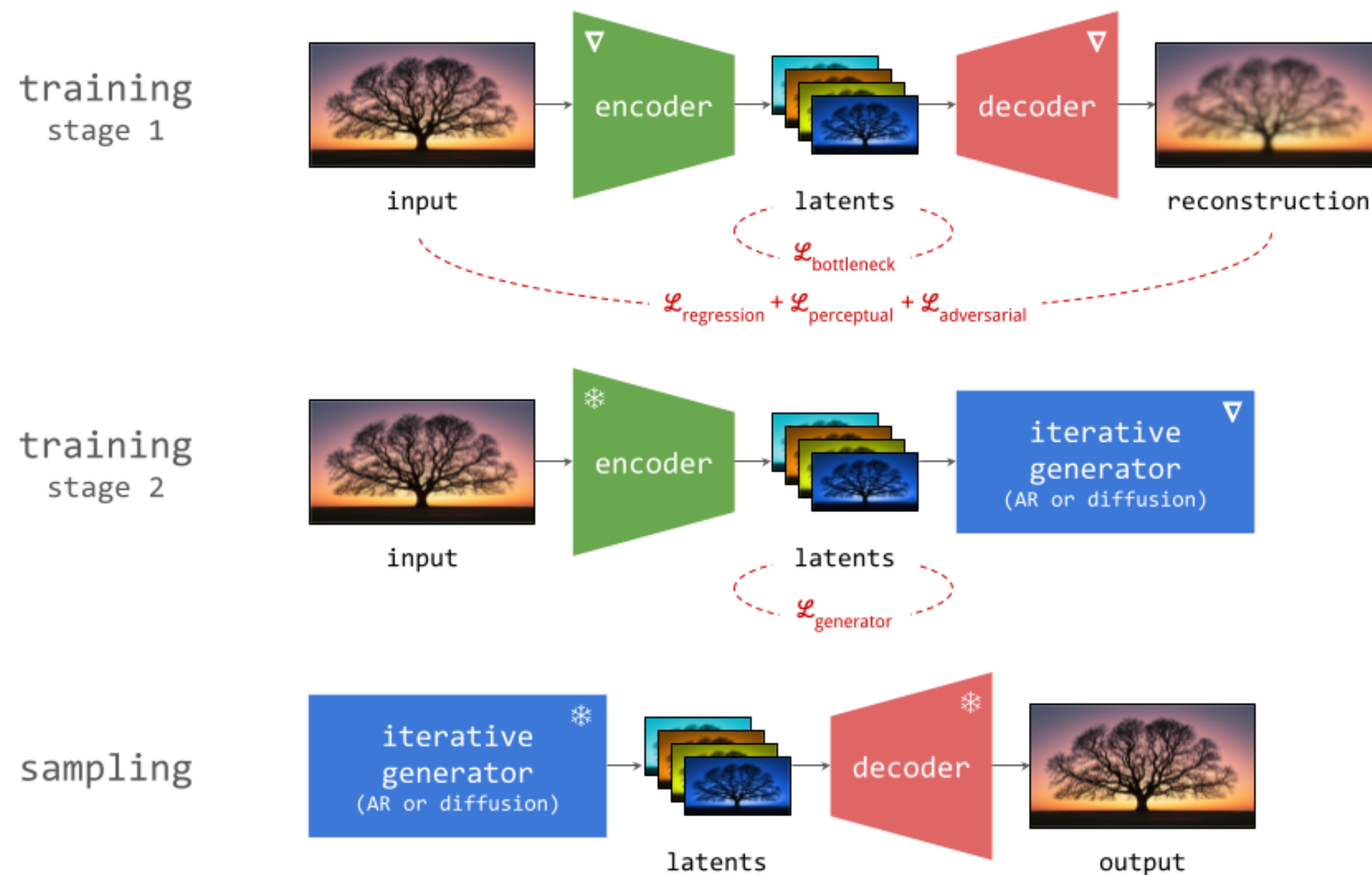
- *Only reconstruction: blurry reconstructions*
- *reconstruction & perception & adversarial & bottleneck*



Latent Space Generation

Two-stage Training

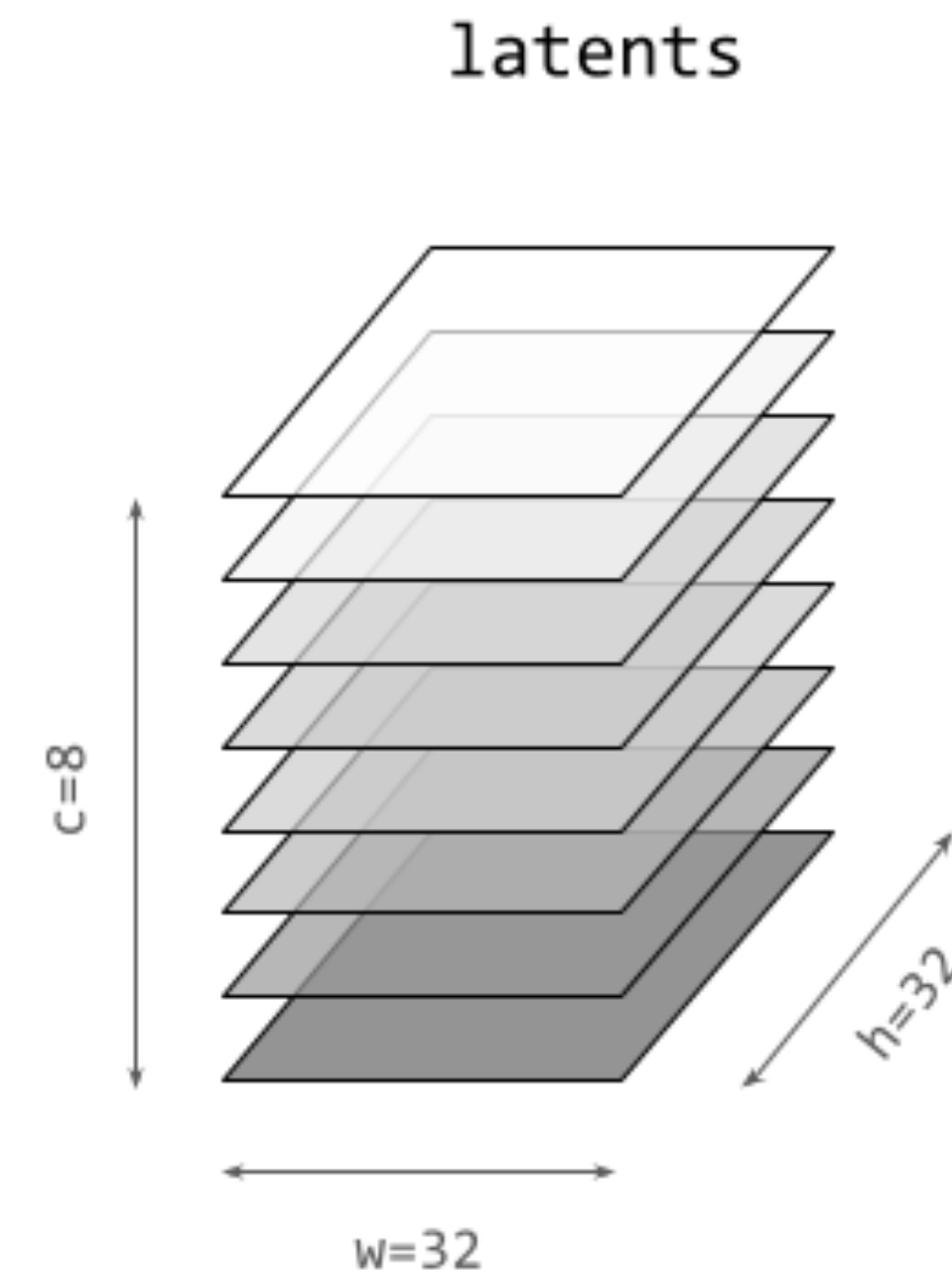
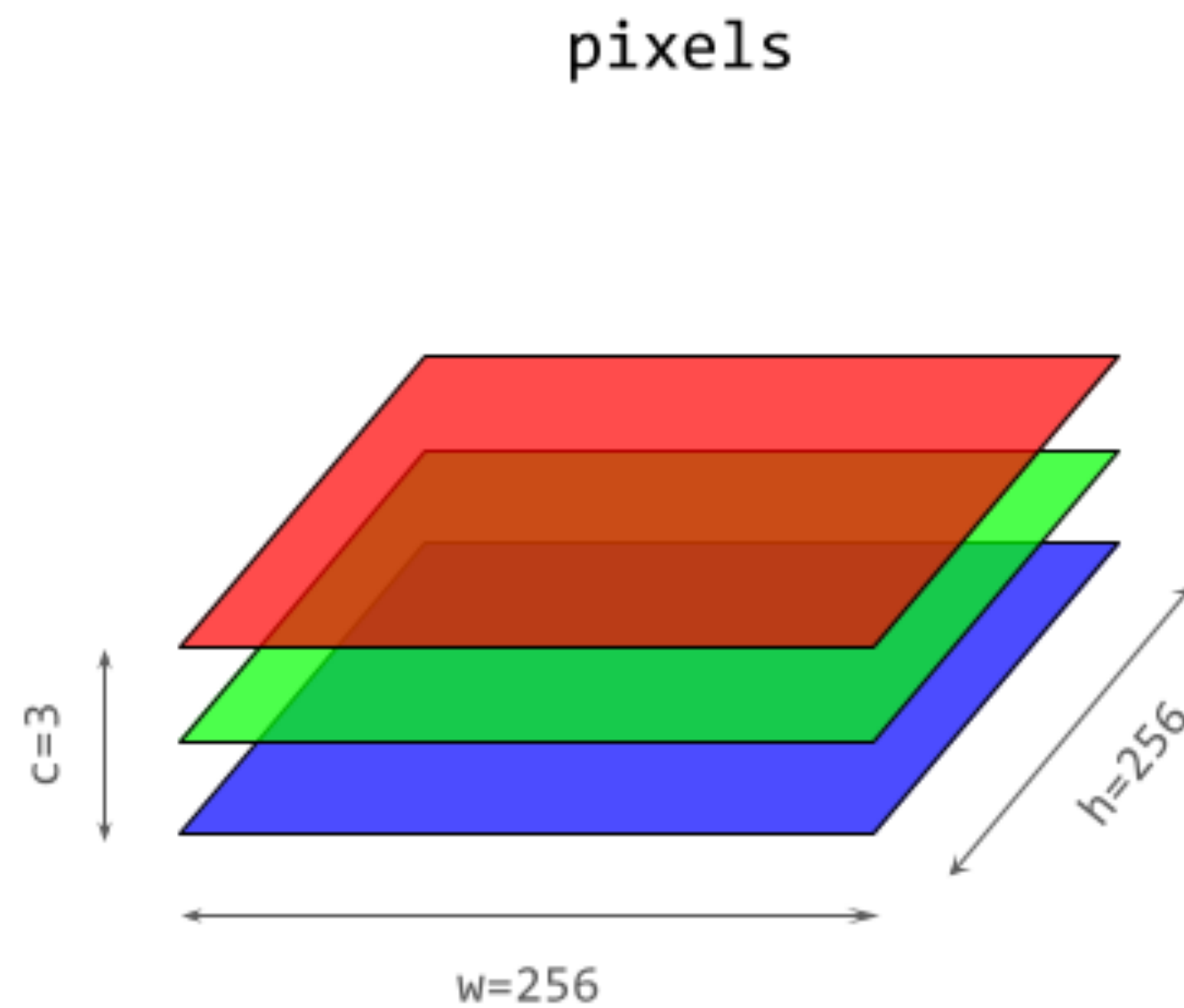
- *Quality of generation is closely related to the **latent space***



Latent Space Generation

Capacity

- *How many bits of information are encoded*
- ***TSR(tensor size reduction factor)***



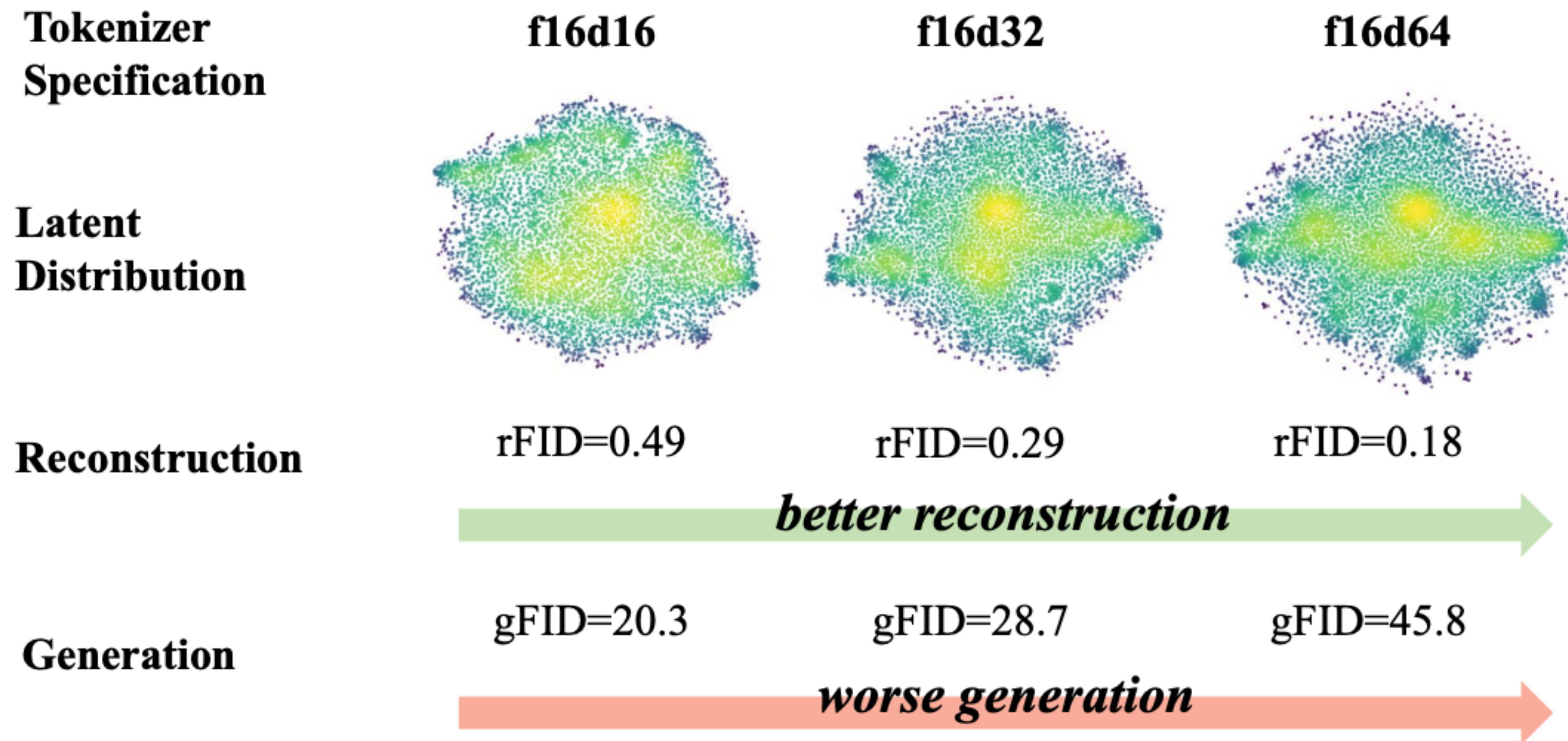
Compression & Representation

Capacity

- *Compression : The tighter, the better*
- *Representation : The more suitable, the better*

Latent Space Generation Capacity

- *Reconstruction vs Generation dilemma*



Latent Space Generation

Curation

- *Which bits from the inout are retained*



Latent Space Generation

Shaping

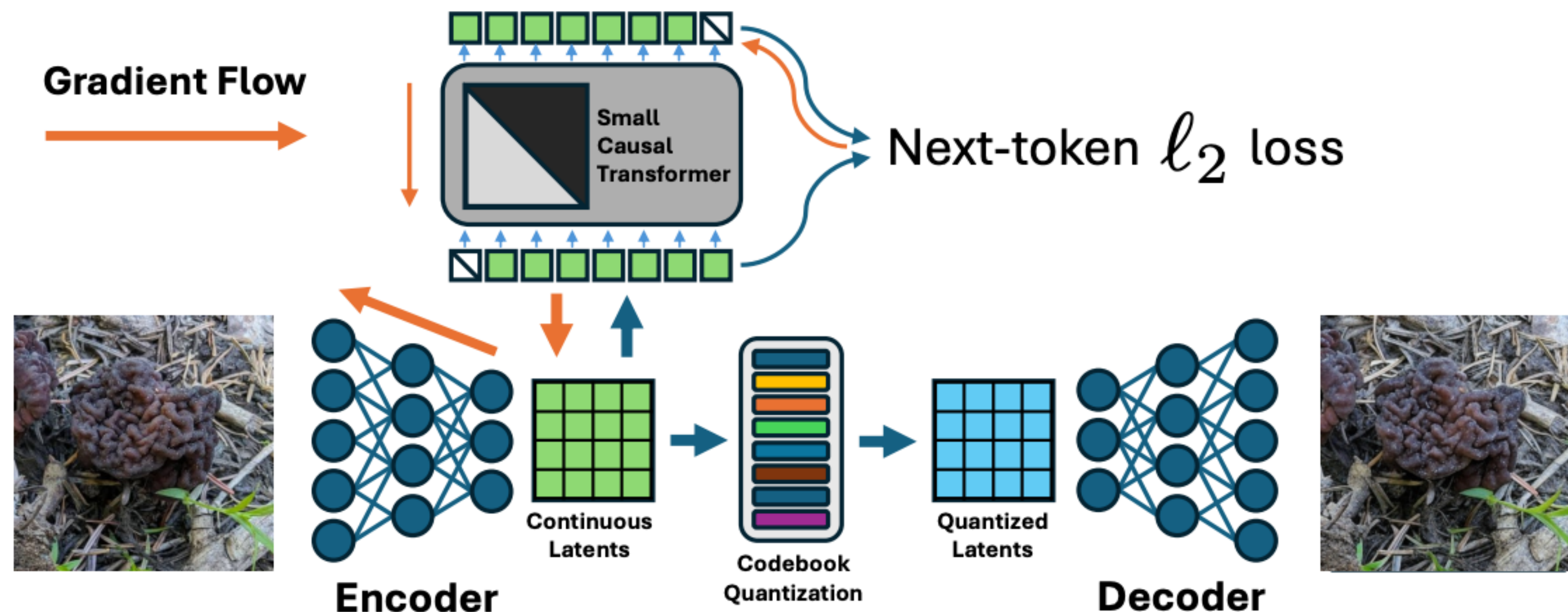
- *How this information is presented*
- ***Scaling&Standardisation***
- ***Continuous & Discrete***
- ***Codebook***

Enhancing Generation Quality

Generative priors

- *Co-train generative model with auto-encoder*
- *Force embeddings to be generation-friendly*

Stage 1: Causally regularized tokenization (CRT)

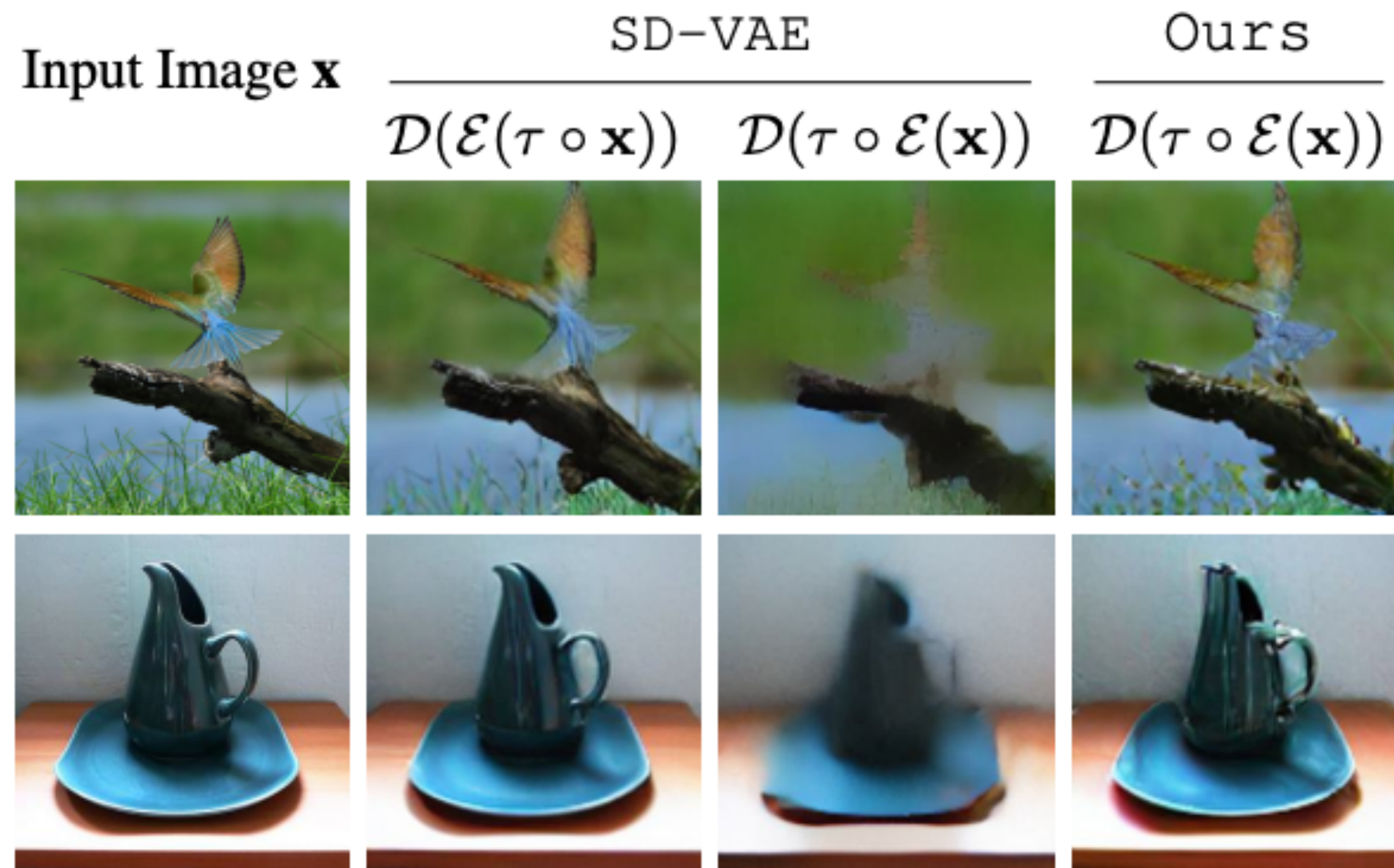


Enhancing Generation Quality

Equivariance

- *Pixel space & Latent space transformation similarly*

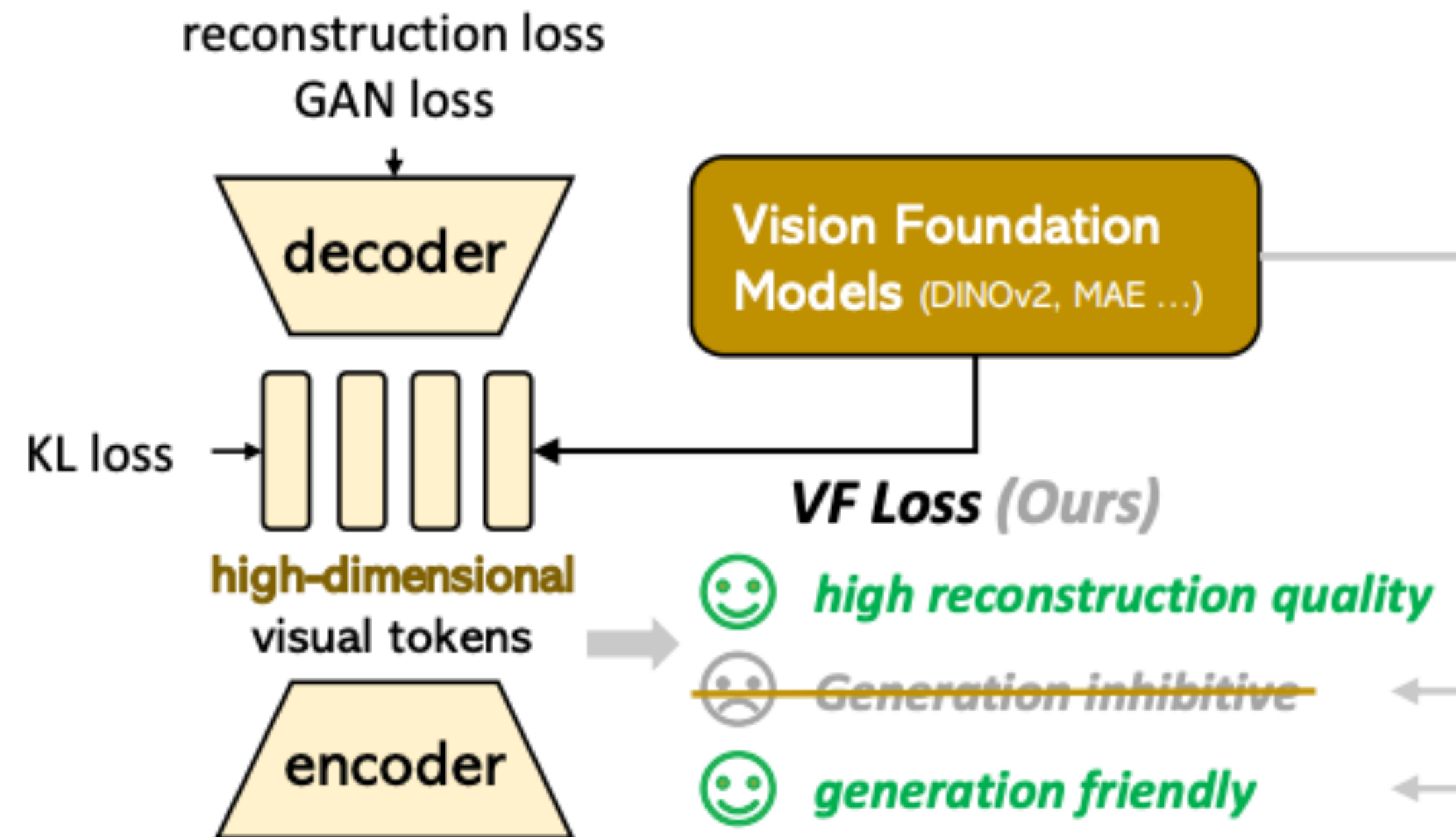
$$\mathcal{L}_{\text{explicit}}(\mathbf{x}) = \|\tau \circ \mathcal{E}(\mathbf{x}) - \mathcal{E}(\tau \circ \mathbf{x})\|_2^2,$$



Enhancing Generation Quality

Pre-trained representations

- *Align latent space with SSL models*

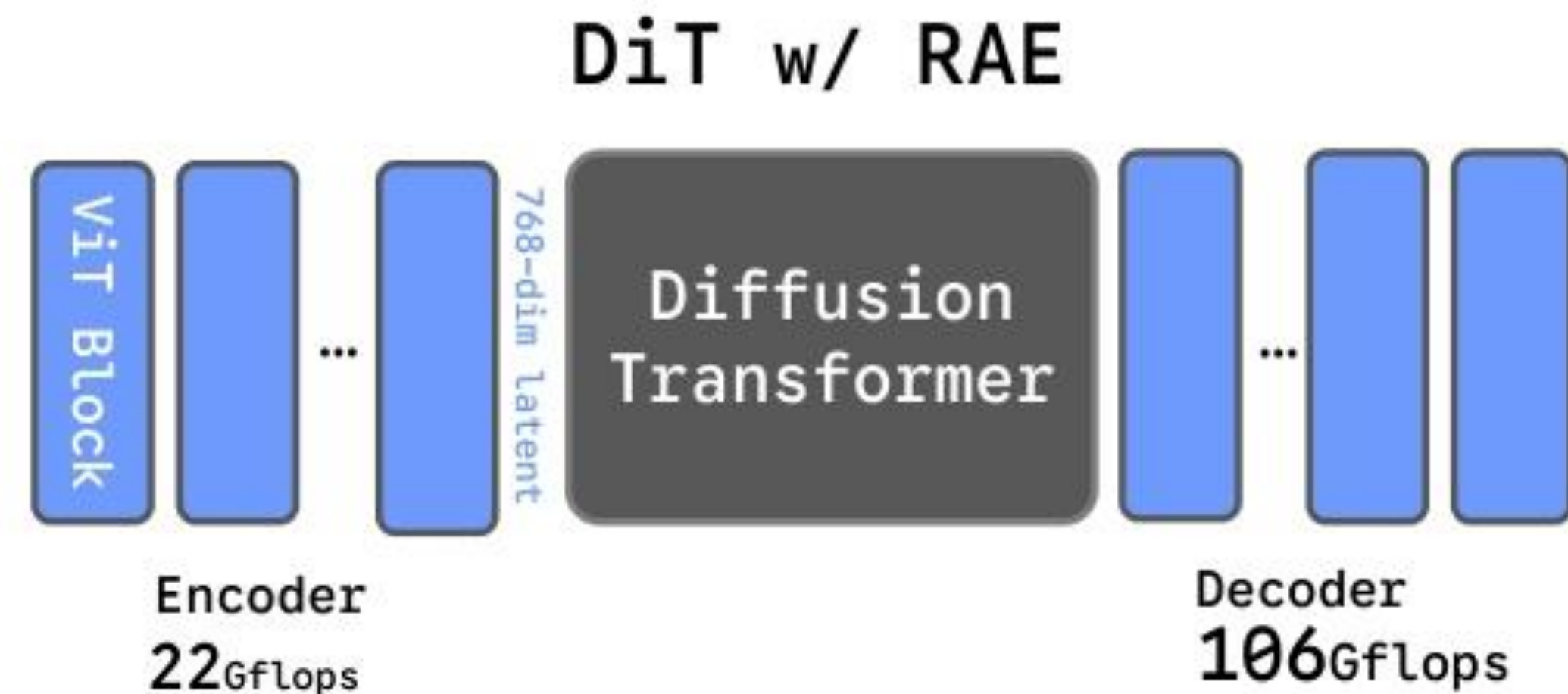
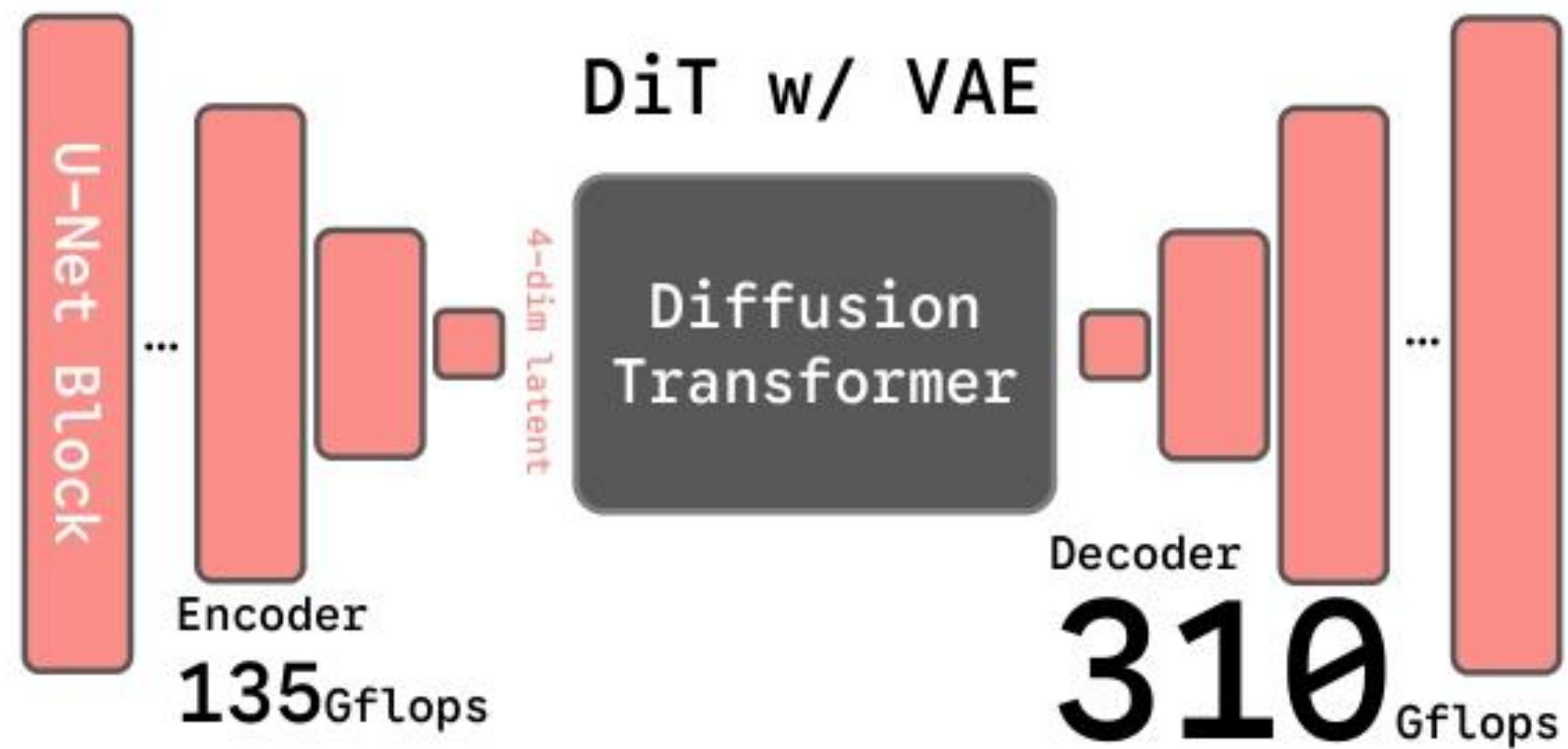


Previous Assumptions

- *SSL Models as Encoder:*
- *Encoder trained to capture semantics are unsuited for faithful reconstruction ?*
- *Diffusion models perform poorly in high-dimensional latent spaces ?*

Diffusion Transformers With Representation Autoencoders

RAE



Reconstruction

$$z = E(x), \hat{x} = D(z)$$

$$\mathcal{L}_{rec}(x) = \omega_L \text{LPIPS}(\hat{x}, x) + \text{L1}(\hat{x}, x) + \omega_G \lambda \text{GAN}(\hat{x}, x)$$

Model	rFID
DINOv2-B	0.49
SigLIP2-B	0.53
MAE-B	0.16
SD-VAE	0.62

(a) **Encoder choice.** All encoders outperform SD-VAE.

Decoder	rFID	GFLOPs
ViT-B	0.58	22.2
ViT-L	0.50	78.1
ViT-XL	0.49	106.7
SD-VAE	0.62	310.4

(b) **Larger decoders** improve rFID while remaining much more efficient than VAEs.

Encoder	rFID
DINOv2-S	0.52
DINOv2-B	0.49
DINOv2-L	0.52

(c) **Encoder scaling.** rFID is stable across RAE sizes.

Model	Top-1 Acc.
DINOv2-B	84.5
SigLIP2-B	79.1
MAE-B	68.0
SD-VAE	8.0

(d) **Representation quality.** RAEs have much higher linear probing accuracy than VAEs.

Generation

Diffusion Transformers

- *Suboptimal design for diffusion transformers (High Dimension)*

- *Suboptimal noise scheduling*

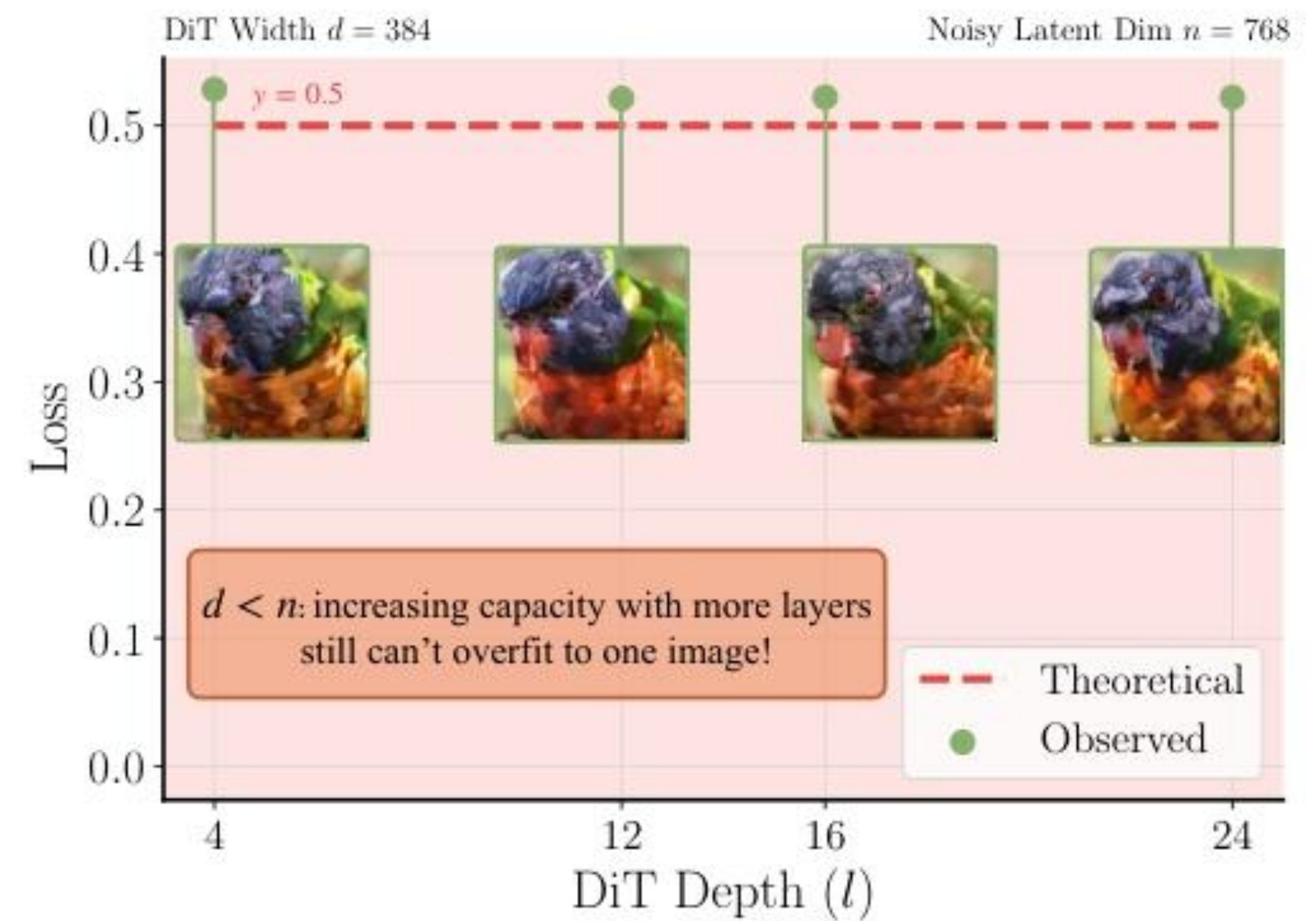
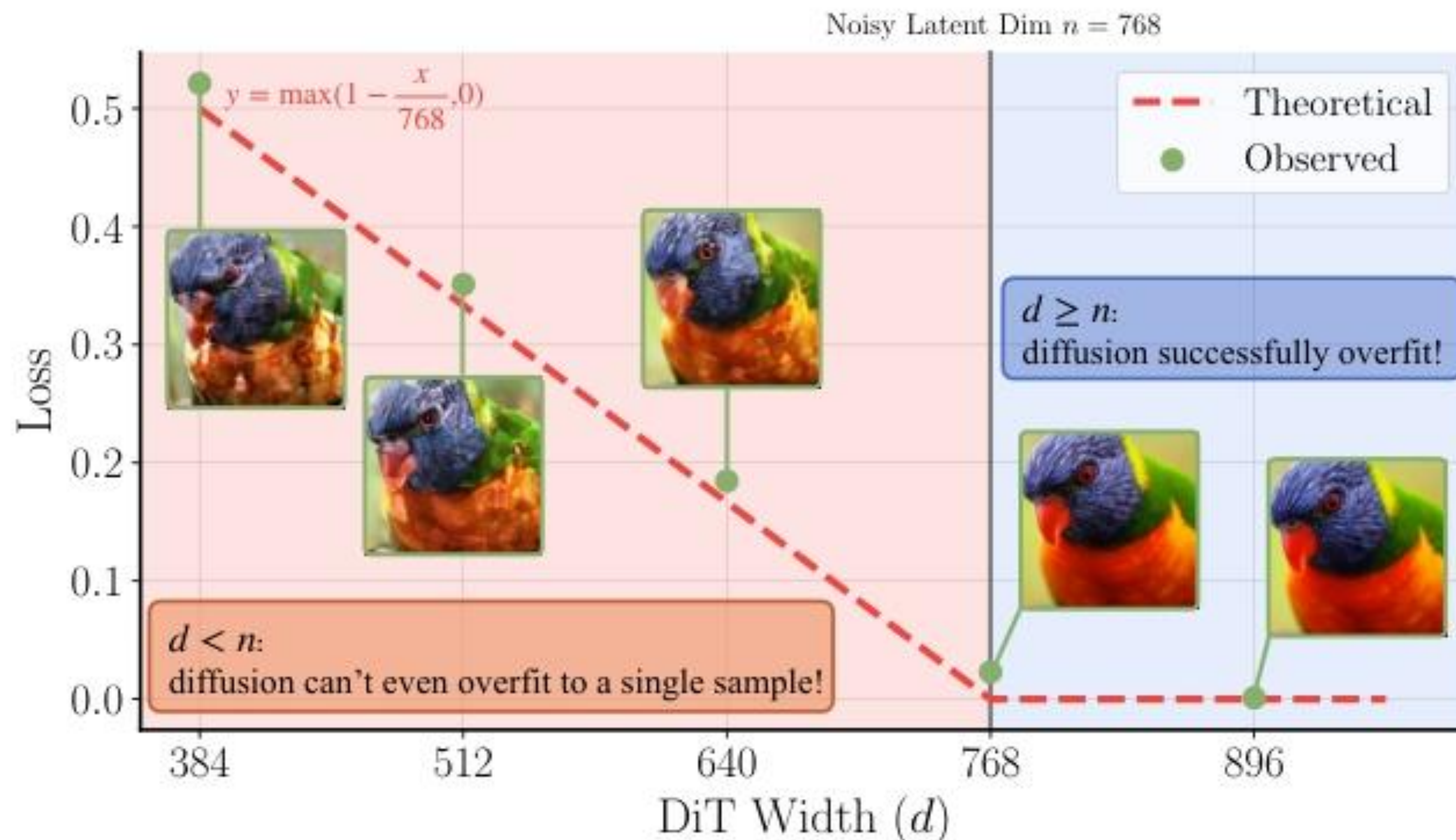
	RAE	SD-VAE
DiT-S	215.76	51.74
DiT-XL	23.08	7.13

- *Diffusion generates noisy latents (VAE decoder trained on noisy latents while RAE decoder on clean latents)*

Generation

Suboptimal design for diffusion transformers

- *Overfit a single image : **DiT width matters** , $d > n$*
- *Injecting noise -> diffusing data into a full-rank manifold*



Generation

Suboptimal noise scheduling

- *Increasing resolution/channel reduces information corruption*
- *Time schedule shifting*

$$t_m = \frac{\alpha t_n}{1 + (\alpha - 1)t_n} \text{ where } \alpha = \sqrt{m/n}$$

gFID	
w/o shift	23.08
w/ shift	4.81

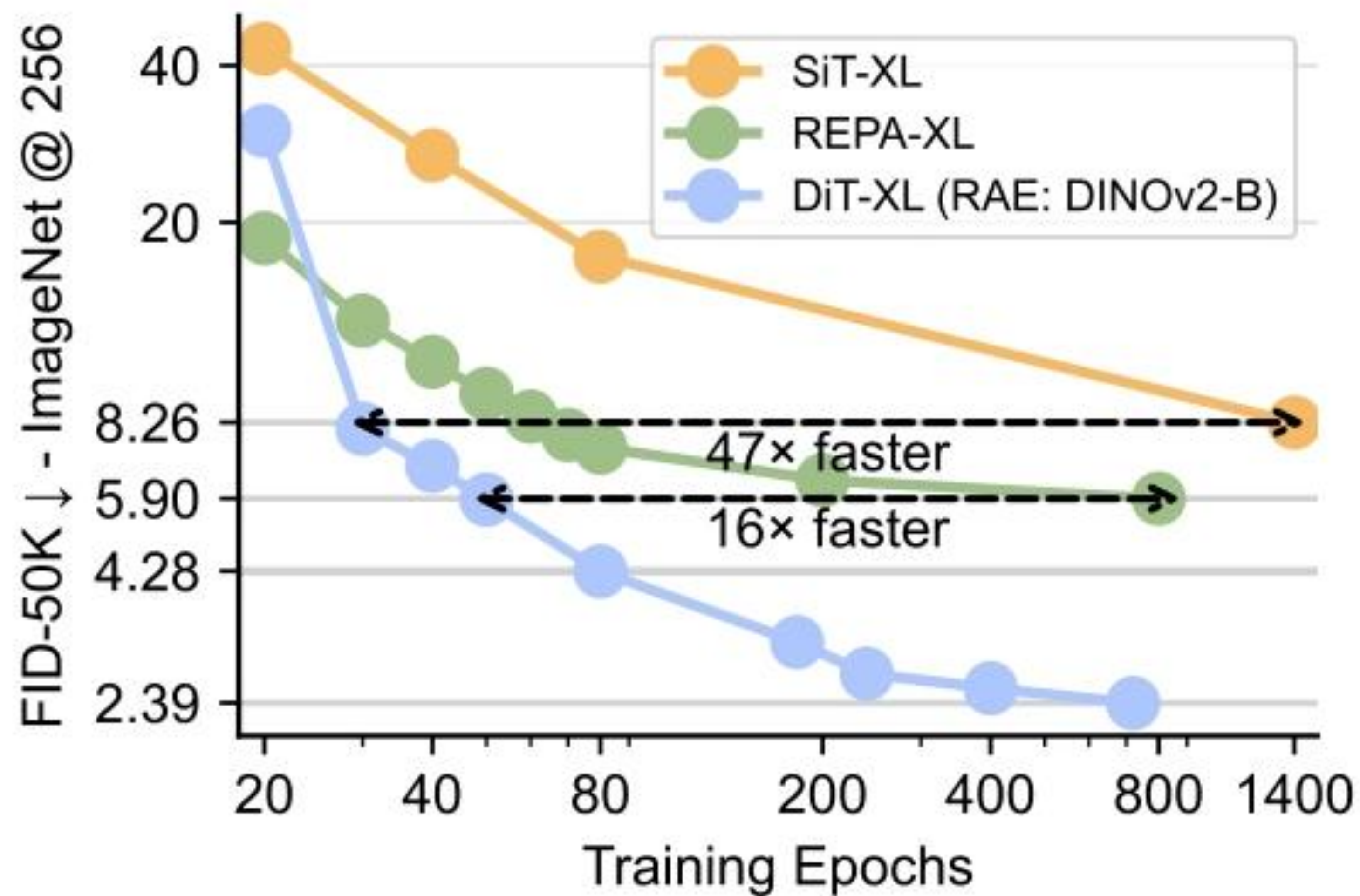
Generation

Noise-augmented decoding

- *VAE trained on continuous space with noise*
- *RAE trained on clean discrete space*
- *Add noise during the training of decoder*

	gFID	rFID
$\mathbf{z} \sim p(\mathbf{z})$	4.81	0.49
$\mathbf{z} \sim p_{\mathbf{n}}(\mathbf{z})$	4.28	0.57

Generation

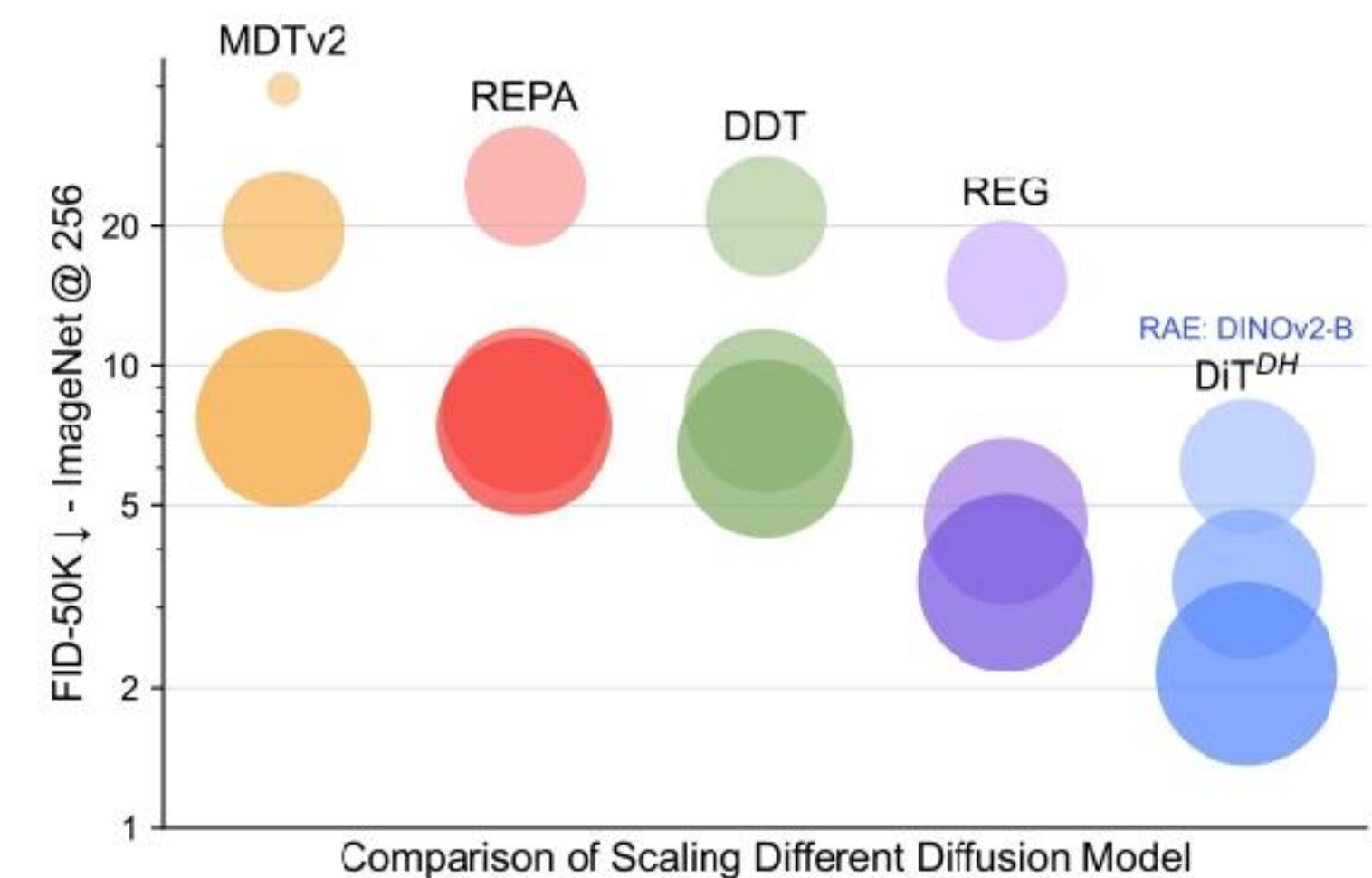
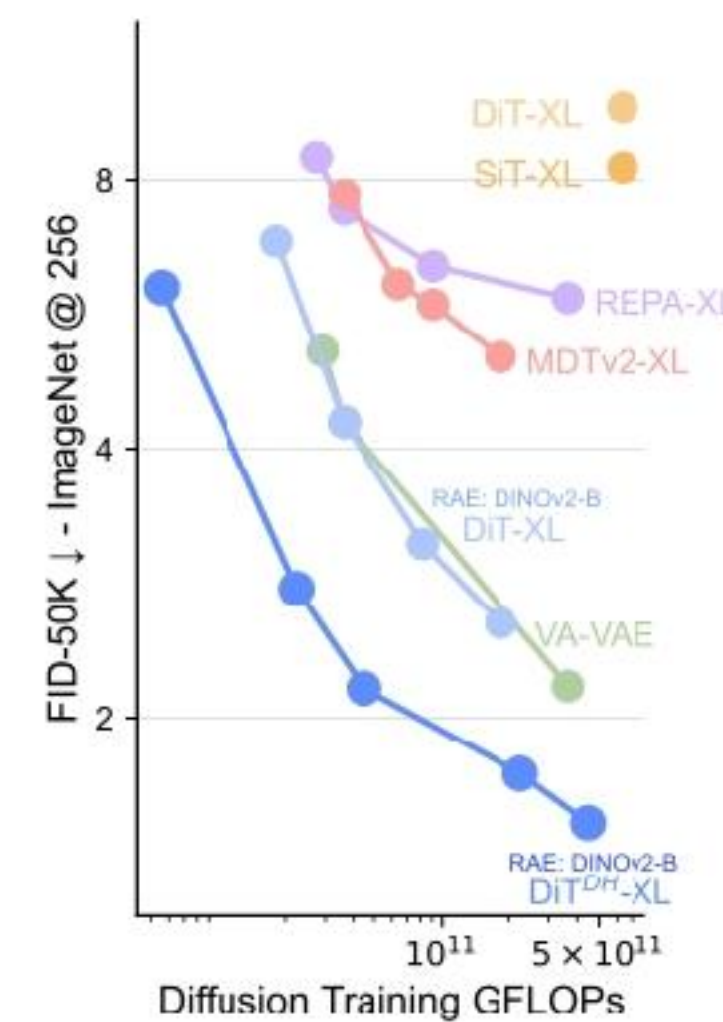
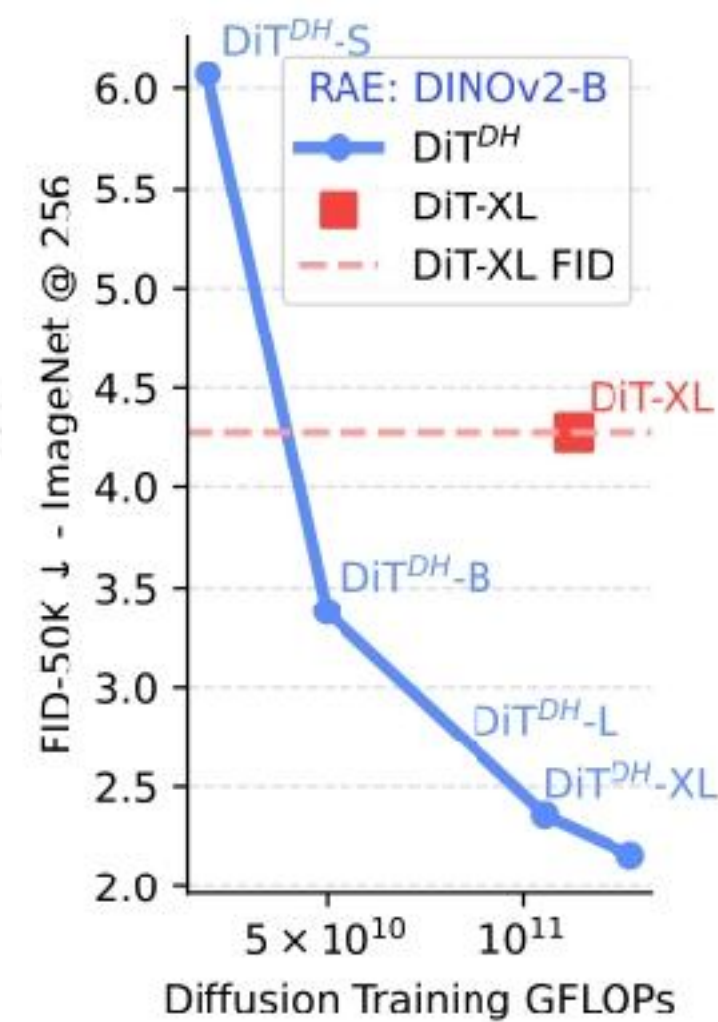
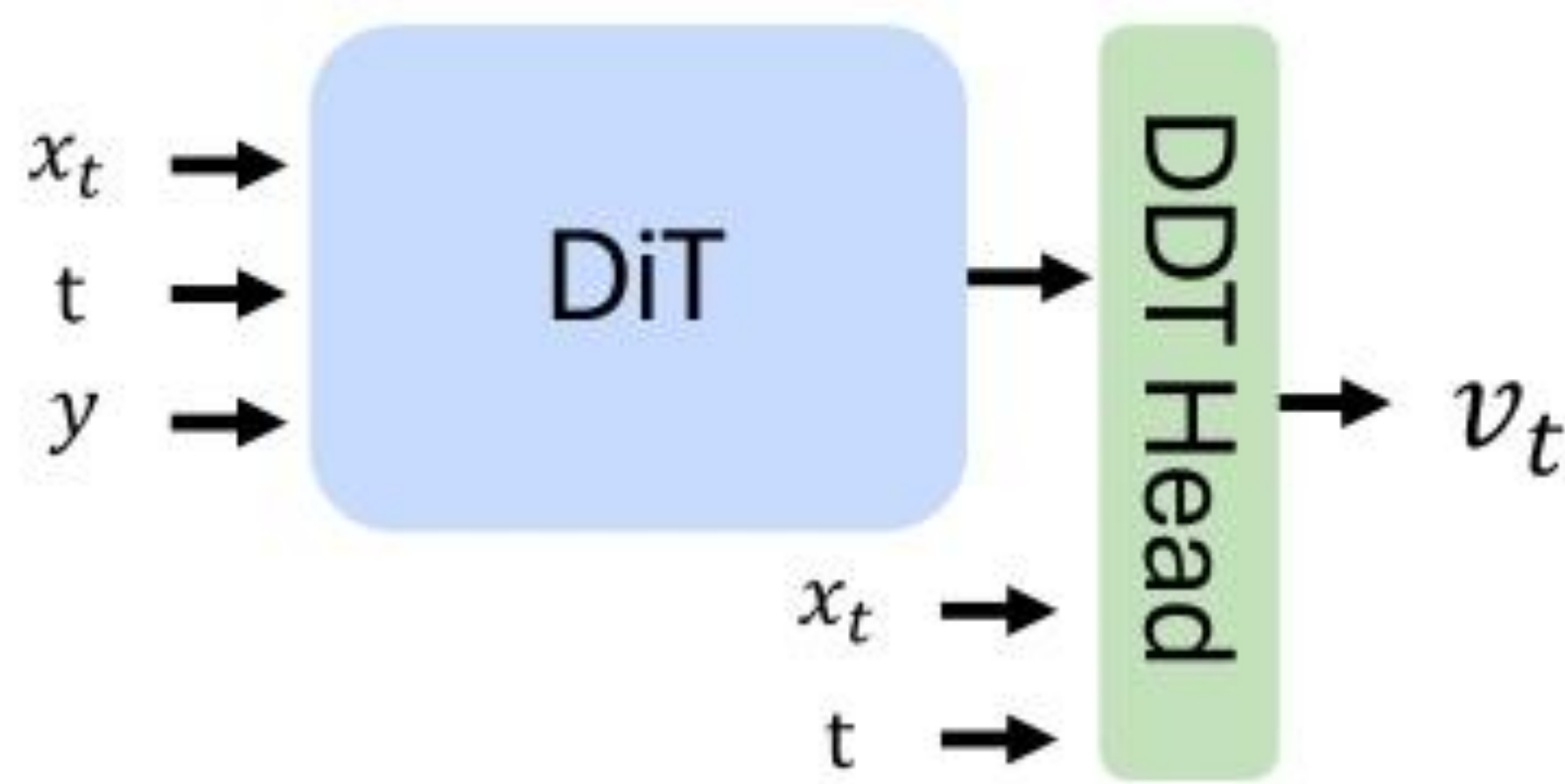


Improving scalability

DiT-DH

- *High dimension : computationally expensive*
- *Wide but shallow head*
- *Converge faster & performs better*

Model	DINOv2		
	S	B	L
DiT-XL	3.50	4.28	6.09
DiT ^{DH} -XL	2.42	2.16	2.73



Results

Method	Epochs	#Params	Generation@256 w/o guidance				Generation@256 w/ guidance			
			gFID↓	IS↑	Prec.↑	Rec.↑	gFID↓	IS↑	Prec.↑	Rec.↑
<i>Autoregressive</i>										
VAR (Tian et al., 2024)	350	2.0B	1.92	323.1	0.82	0.59	1.73	350.2	0.82	0.60
MAR (Li et al., 2024b)	800	943M	2.35	227.8	0.79	0.62	1.55	303.7	0.81	0.62
xAR (Ren et al., 2025)	800	1.1B	-	-	-	-	1.24	301.6	0.83	0.64
<i>Pixel Diffusion</i>										
ADM (Dhariwal & Nichol, 2021)	400	554M	10.94	101.0	0.69	0.63	3.94	215.8	0.83	0.53
RIN (Jabri et al., 2023)	480	410M	3.42	182.0	-	-	-	-	-	-
PixelFlow (Chen et al., 2025e)	320	677M	-	-	-	-	1.98	282.1	0.81	0.60
PixNerd (Wang et al., 2025b)	160	700M	-	-	-	-	2.15	297.0	0.79	0.59
SiD2 (Hooeboom et al., 2025)	1280	-	-	-	-	-	1.38	-	-	-
<i>Latent Diffusion with VAE</i>										
DiT (Peebles & Xie, 2023)	1400	675M	9.62	121.5	0.67	0.67	2.27	278.2	0.83	0.57
MaskDiT (Zheng et al.)	1600	675M	5.69	177.9	0.74	0.60	2.28	276.6	0.80	0.61
SiT (Ma et al., 2024)	1400	675M	8.61	131.7	0.68	0.67	2.06	270.3	0.82	0.59
MDTv2 (Gao et al., 2023)	1080	675M	-	-	-	-	1.58	314.7	0.79	0.65
VA-VAE (Yao et al., 2025)	80	675M	4.29	-	-	-	-	-	-	-
	800		2.17	205.6	0.77	0.65	1.35	295.3	0.79	0.65
REPA (Yu et al., 2025)	80	675M	7.90	122.6	0.70	0.65	-	-	-	-
	800		5.78	158.3	0.70	0.68	1.29	306.3	0.79	0.64
DDT (Wang et al., 2025c)	80	675M	6.62	135.2	0.69	0.67	1.52	263.7	0.78	0.63
	400		6.27	154.7	0.68	0.69	1.26	310.6	0.79	0.65
REPA-E (Leng et al., 2025)	80	675M	3.46	159.8	0.77	0.63	1.67	266.3	0.80	0.63
	800		1.70	217.3	0.77	0.66	1.15	304.0	0.79	0.66
<i>Latent Diffusion with RAE (Ours)</i>										
DiT-XL (DINOv2-S)	800	676M	1.87	209.7	0.80	0.63	1.41	309.4	0.80	0.63
DiT ^{DH} -XL (DINOv2-B)	20	839M	3.71	198.7	0.86	0.50	-	-	-	-
	80		2.16	214.8	0.82	0.59	-	-	-	-
	800		1.51	242.9	0.79	0.63	1.13	262.6	0.78	0.67